

Assignment -1

(Deadline : 2026 - 06 - 25)

1. Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Process	Process
P1	5	4
P2	3	1
P3	1	2
P4	7	2
P5	4	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

- Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
 - What is the turnaround time of each process for each of the scheduling algorithms in part a?
 - What is the waiting time of each process for each of these scheduling algorithms?
 - Which of the algorithms results in the minimum average waiting time (over all processes)?
2. The following processes are being scheduled using a preemptive, priority-based, round-robin scheduling algorithm.

Process	Priority	Burst	Arrival
P1	8	15	0
P2	3	20	0
P3	4	20	20
P4	4	20	25
P5	5	5	45
P6	5	15	55

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. The scheduler will execute the highest-priority process. For processes with the same priority, a round-robin scheduler will be used with a time quantum of 10 units. If

a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

a) Show the scheduling order of the processes using a Gantt chart.

b) What is the turnaround time for each process?

c) What is the waiting time for each process?

3. Consider below table of three process under multi-level feedback queue scheduling. MFQ scheduler with three queue Q1 (RR with time quantum=8), Q2 (RR with time quantum=16) and Q3 (FCFS).

Process	Arrival Time	Burst Time
P1	0	36
P2	16	20
P3	20	12

Draw Gantt chart to illustrate their execution and calculate average waiting and Turnaround time also.

4. Consider a System with 5 concurrent processes (P0,P1,P2,P3,P4) and 4 resource types (R0,R1,R2,R3).The number of instance of each resource type in the system are 6,4,4,2 respectively. Allocation table and maximum claim table are as follows

	R0	R1	R2	R3
P0	3	2	1	1
P1	1	2	0	2
P2	1	1	2	0
P3	3	2	1	0
P4	2	1	0	1

Maximum Claim

	R0	R1	R2	R3
P0	2	0	1	1
P1	1	1	0	0
P2	1	1	0	0
P3	1	0	1	0
P4	0	1	0	1

Allocation

Is the state safe? If so, Show the safe execution of processes

5. Consider the following System:

process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Answer the following questions using Banker's algorithm:

- What will be the contents of matrix need?
 - Draw the safe sequence?
 - If the request from P1 Arrives for (0, 4, 2, 0) can the request be granted immediately?
6. Given three process A, B, and C and three resources X, Y, and Z are following events happen in the sequence as given:
- A request X
 - B request Y
 - C request Z
 - A request Y
 - B request Z
 - C request X
 - C request Y

Assume that requested resource should be always be allocated to request process if it is available. Draw the resource allocation graph for the sequence and also mention whether it is deadlock? If it is how to recover the deadlock?